GMR Institute of Technology

Lecture 26
Structure Modeling (2 to 4 Decoder)

Contents of the lecture

(1) Structure model of 2 to 4 Decoder

(2) Test bench of 2 to 4 Decoder

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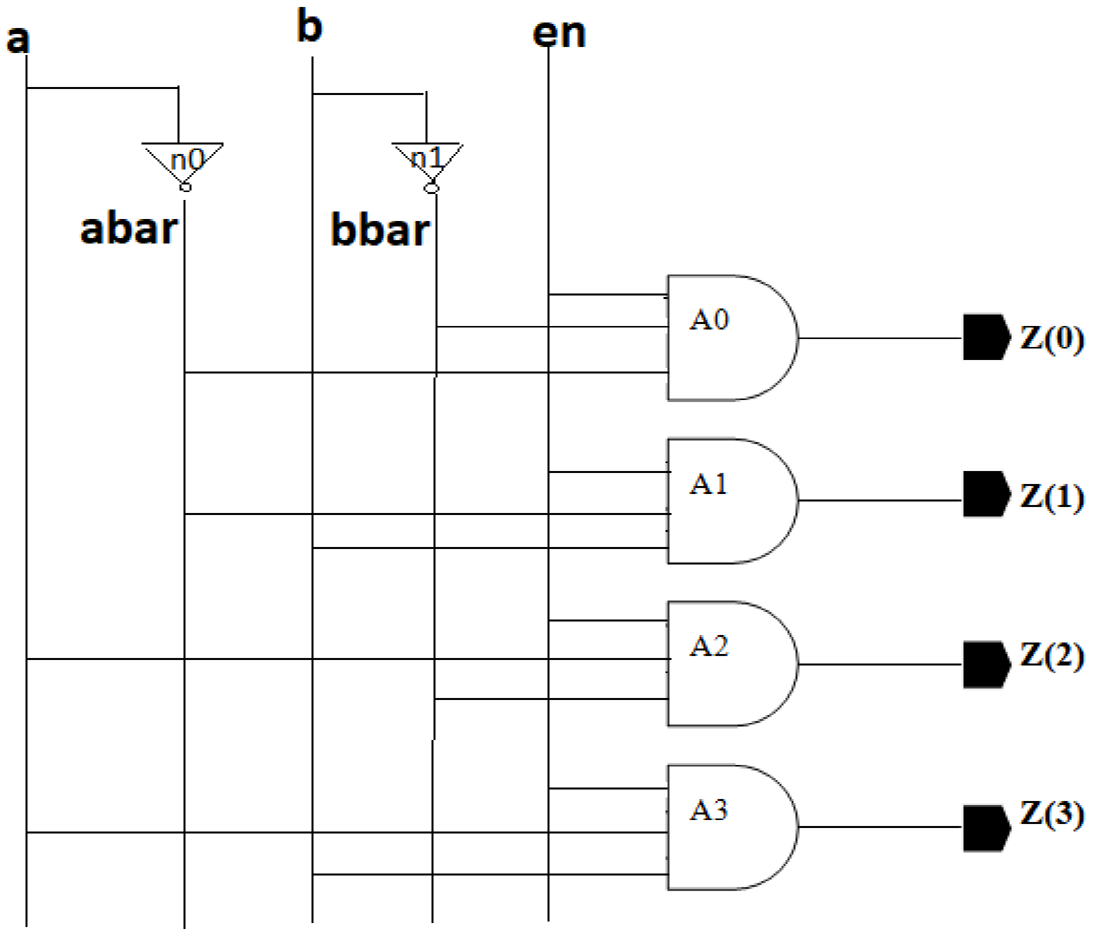
```
module decoder_2_to_4 (a, b, en,z);
input a, b, en;
output [0:3]z;
wire abar, bbar;
not n0 (abar, a);
not n1 (bbar, b);
and a0 (z[0], abar, bbar, en);
and a1 (z[1], abar, b, en);
and a2 (z[2], a, bbar, en);
and a3 (z[3], a, b, en);
endmodule
```



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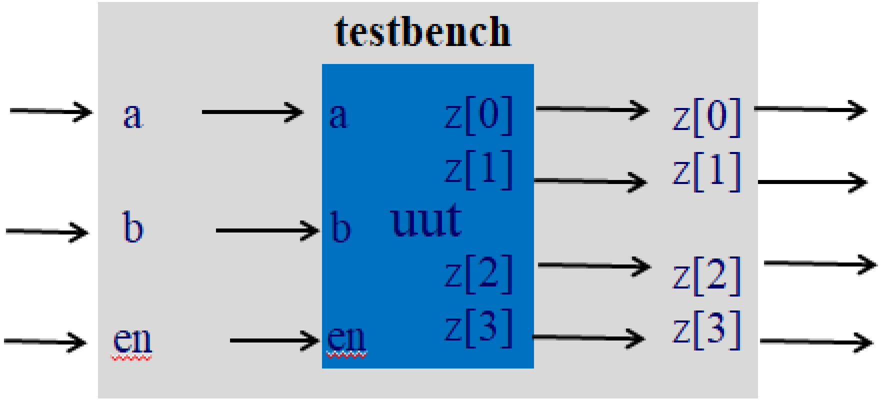
and a1 (z[1], abar, b, en);
and a2 (z[2], a, bbar, en);
and a3 (z[3], a, b, en);
endmodule
```

```
module  testbench;
reg a;reg b;reg  en;wire  [0:3]z;
decoder_2_to_4 uut (.a(a),.b(b),.en(en),.z(z));
initial
begin
$monitor($time,"a=%b,b=%b,en=%b,z=%b",a,b,en,z);
#5 a=0;b=0;en=1;
#5 a=0;b=1;en=1;
#5 a=1;b=0;en=1;
#5 a=1;b=1;en=1;
#5 a=0;b=0;en=0;
#5 a=0;b=1;en=0;
#5 $finish;
end
endmodule
```



Software Demonstration of Test bench of 2 to 4 Decoder